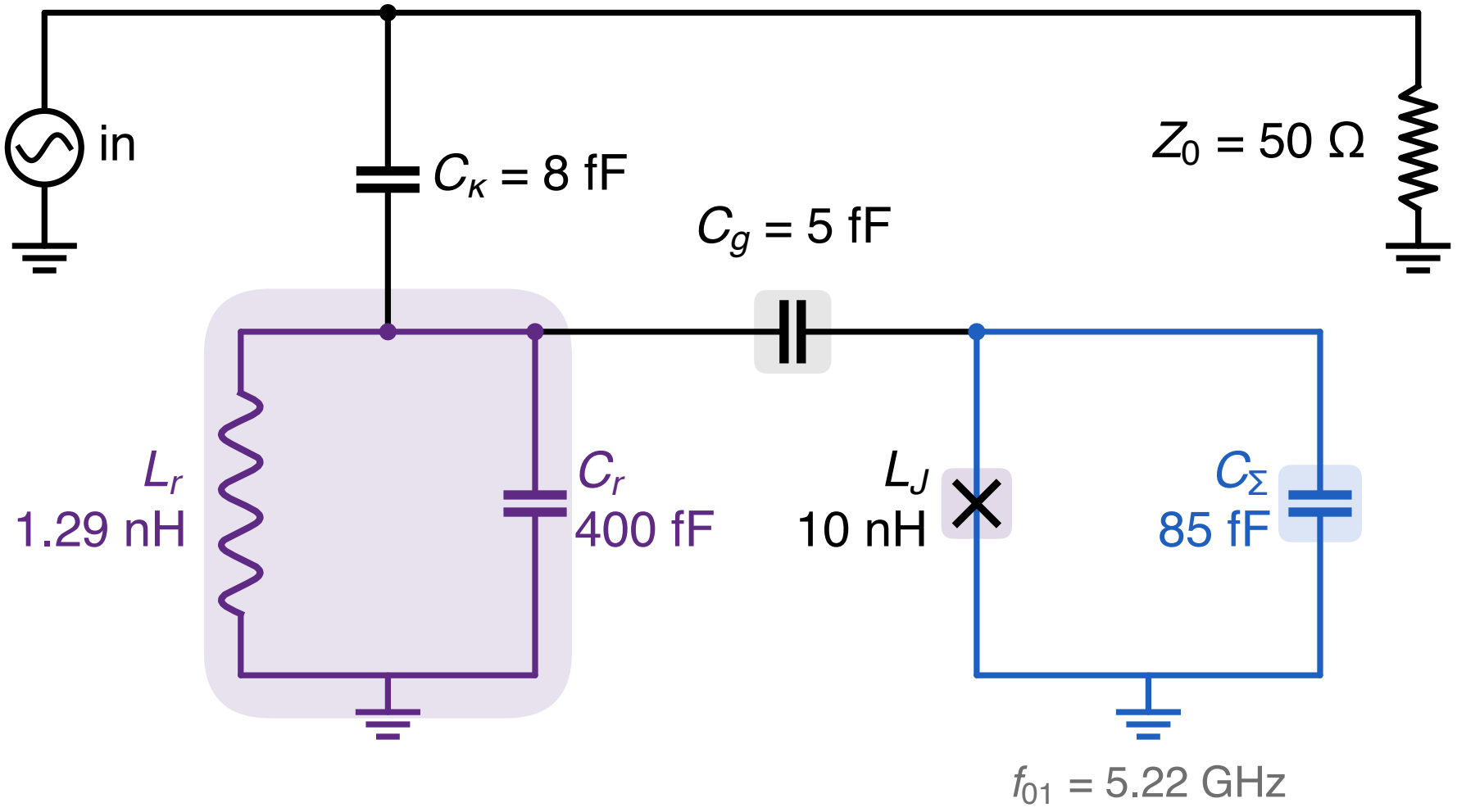

$$\hat{H} = \underbrace{4E_C\hat{n}^2}_{E_C/h = 228\ \text{MHz}} + \underbrace{-E_J\cos\hat{\phi}}_{E_J/h = 16.3\ \text{GHz}} + \underbrace{hf_r\hat{a}^\dagger\hat{a}}_{f_r = 7.00\ \text{GHz}} + \underbrace{hg\hat{n}(\hat{a}+\hat{a}^\dagger)}_{g/h = 82\ \text{MHz}}$$